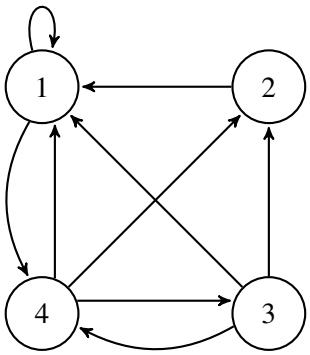
EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 7A

1. PageRank: Computing the Score Vector

Consider the network of four webpages below. An arrow from node j to node i means that page j links to page i.



Each page splits its visitors equally among the pages it links to, so the connectivity matrix is

$$\mathbf{C} = \begin{bmatrix} \frac{1}{2} & 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{3} & 0 \end{bmatrix}.$$

Let $\mathbf{s}[n] = \begin{bmatrix} s_1[n] & s_2[n] & s_3[n] & s_4[n] \end{bmatrix}^\top$ hold the fraction of all visitors on each page at time step n. At every step each visitor clicks a link, so

$$\mathbf{s}[n+1] = \mathbf{C}\,\mathbf{s}[n].$$

(a) Suppose the visitors start out spread evenly over the four pages: $\mathbf{s}[0] = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}^\top$. Compute $\mathbf{s}[1]$. What do its entries sum to, and why?

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(b) The PageRank *score vector* $\mathbf{s}^*$ is the distribution of visitors that no longer changes from one step to the next; this is the *equilibrium state* of the network. Write down the equation that $\mathbf{s}^*$ must satisfy. What does this say about $\mathbf{s}^*$ in terms of the eigenvalues and eigenvectors of $\mathbf{C}$? In terms of a null space?

(c) Find the score vector $\mathbf{s}^*$ for this network.

(d) Rank the four pages from most to least important. Pages 2 and 4 each have exactly two incoming links. Do they receive the same score? Explain what you see.

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2. Diagonalization and PageRank

The PageRank *score vector* $\mathbf{s}^*$ of a network of webpages is the distribution of visitors that no longer changes from one step to the next: $\mathbf{C}\mathbf{s}^* = \mathbf{s}^*$, where $\mathbf{C}$ is the connectivity matrix and the entries of $\mathbf{s}^*$ sum to 1. In other words, $\mathbf{s}^*$ is an eigenvector of $\mathbf{C}$ associated with the eigenvalue $\lambda = 1$. In this problem we use *all* of the eigenvalues and eigenvectors of $\mathbf{C}$ to understand why visitors settle into the score vector, and how quickly.

Consider a network of two webpages with connectivity matrix

$$\mathbf{C} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix},$$

where entry (i, j) is the fraction of page j's visitors who move to page i at each step. If $\mathbf{s}[n]$ holds the fraction of all visitors on each page at step n, then $\mathbf{s}[n+1] = \mathbf{C}\mathbf{s}[n]$.

(a) Draw the graph of this network. Use one node per webpage, and label each arrow with the fraction of visitors who follow it at each step.

(b) Find the eigenvalues λ_1 and λ_2 of $\mathbf{C}$, and an eigenvector $\mathbf{v}_1$, $\mathbf{v}_2$ associated with each. What is the PageRank score vector $\mathbf{s}^*$?

(c) Let $\mathbf{V} = \begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2\end{bmatrix}$ be the matrix whose columns are the eigenvectors, and let $\mathbf{\Lambda} = \begin{bmatrix}\lambda_1 & 0 \\ 0 & \lambda_2\end{bmatrix}$. Verify that $\mathbf{CV} = \mathbf{V\Lambda}$. Then compute $\mathbf{V}^{-1}$ and write $\mathbf{C}$ in its *diagonalized* form $\mathbf{C} = \mathbf{V\Lambda V}^{-1}$.

(d) After n steps the visitors are distributed as $\mathbf{s}[n] = \mathbf{C}^n\mathbf{s}[0]$. Use the diagonalized form to show that $\mathbf{C}^n = \mathbf{V\Lambda}^n\mathbf{V}^{-1}$, and use this to write each entry of $\mathbf{C}^n$ in terms of n.

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(e) What does $\mathbf{C}^n$ approach as $n \to \infty$? Use your answer to show that $\mathbf{s}[n]$ converges to the score vector $\mathbf{s}^*$ for *every* starting distribution $\mathbf{s}[0] = \begin{bmatrix} a \\ b \end{bmatrix}$, where $\mathbf{s}[0]$ is a nonnegative vector ($a, b \geq 0$) with $a + b = 1$.

(f) Now consider a different network of two webpages in which each page links only to the other, so that its connectivity matrix is

$$\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The eigenvalues of $\mathbf{P}$ are $\lambda_1 = 1$ and $\lambda_2 = -1$. Diagonalize $\mathbf{P}$ and use the result to write each entry of $\mathbf{P}^n$ in terms of n. Does $\mathbf{P}^n$ approach a limit as $n \to \infty$? If every visitor starts on page 1, do the visitors settle into a steady distribution? Compare with what happened for $\mathbf{C}$.

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